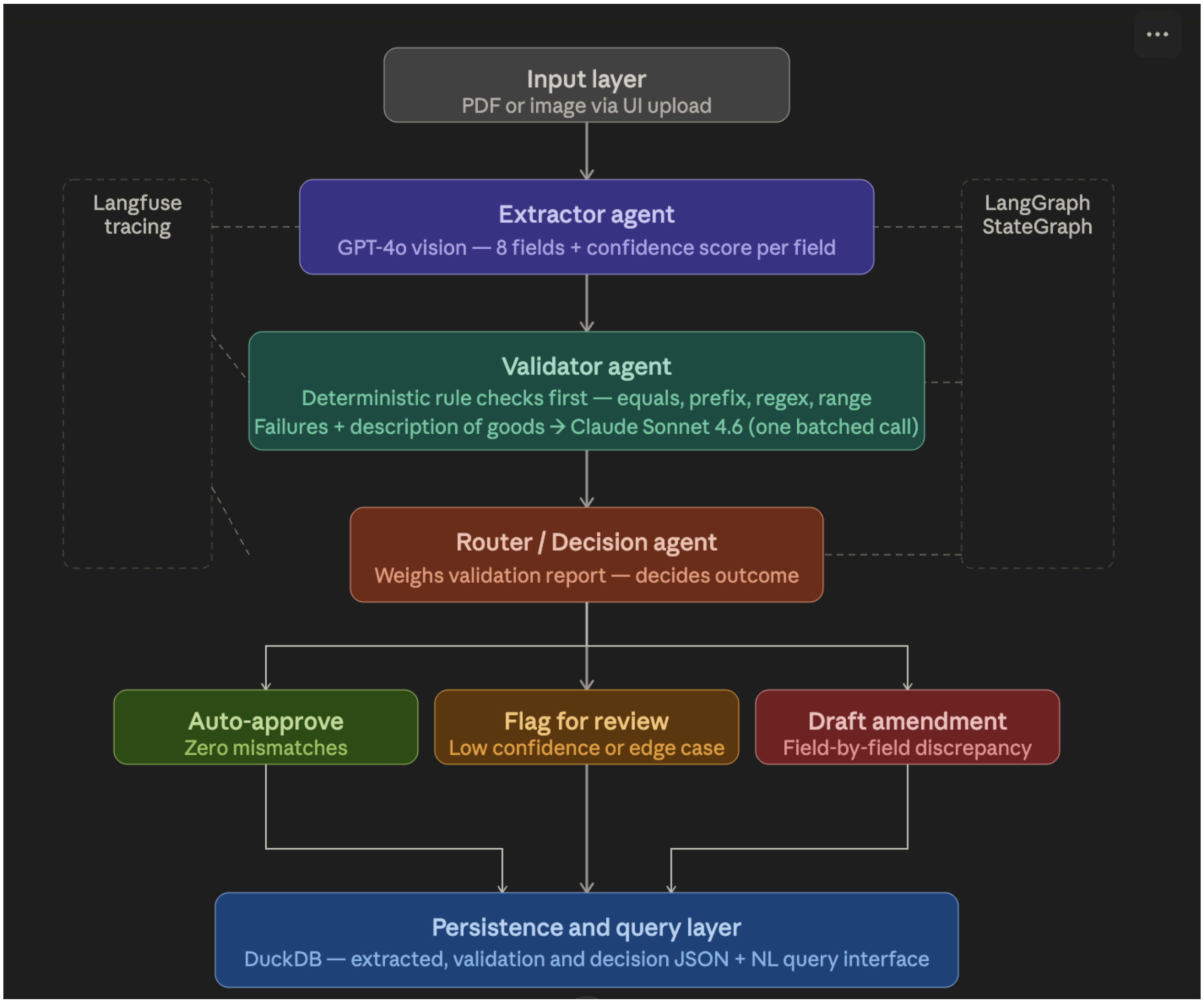
Technical Write-up: Multi-Agent Trade Pipeline

This document outlines the architecture, failure handling, and operational metrics for the Nova Trade Document Pipeline POC. The system is designed to transform unstructured trade documents into validated, actionable data for Cargo Group (CG) operators.

System Architecture

The pipeline follows a sequential multi-agent pattern to ensure high precision and clear separation of concerns.

- **Input Layer:** Trade documents (PDF/Images) are ingested via the UI.
- **Extractor Agent:** Utilizes a vision-capable LLM to map structured JSON schema (consignee, HS code, etc.) with field-level confidence scores.
- **Validator Agent:** Cross-references extracted data against a predefined customer rule set, identifying matches, mismatches, or uncertainties.
- **Router / Decision Agent:** Analyzes validation results to determine the next workflow step: auto-approval, human review flag, or automated amendment drafting.
- **Persistence & Query Layer:** Finalized states are stored in a queryable database (SQLite for POC), accessible via a natural-language query interface.



Failure Mode Analysis

Over-Reliance on Brittle Rule-Based Validation Initially, the Validator Agent used a hybrid approach: rigid, rule-based matching for structured fields (like dates, numbers) and LLM semantic matching for descriptive fields (like product descriptions). I realized that the rule-based component was brittle, requiring constant updates and failing to catch nuanced mismatches. **Resolution:** We pivoted to a "Rule-First, LLM-Fallback" model. Fields that fail the initial rule-based validation are now passed to the LLM for a semantic double-check, alongside the field's expected value. This ensures we catch edge cases that break simple rules without sacrificing the speed of the initial validation layer.

Observability & Production Monitoring

Shipment Tracing

Each shipment is assigned a unique `Shipment_ID` at the moment of ingestion. We use distributed tracing to follow this ID through every agent hop. If a validation fails, the trace shows exactly which agent (Extractor or Validator) caused the bottleneck or error.

Performance Dashboard

Metric	Description
Extraction Accuracy	Percentage of fields where agent confidence matches human verification.
STP Rate	Straight-Through Processing: shipments auto-approved without CG intervention.
Agent Latency	P95 time for each agent (Vision extraction is typically the bottleneck).
Cost per Doc	Real-time tracking of token usage across different LLM models.
Human Override Rate	Frequency at which CG operators edit or reject agent-drafted amendments.

Cost & Latency Profile

For one PDF document with single page data, the cost observed was -

Back-of-Envelope Cost

The primary cost driver is the vision-capable LLM used for extraction.

- **Estimated Cost:** ~$0.02 to $0.15 per document, depending on page count and token density.
- **Cost Control:** Future iterations will use a hybrid layer—standardized layouts will be processed via lightweight OCR/NER, reserving expensive Vision LLMs for unknown or messy layouts.

Latency Bottlenecks

For the same run on a single page PDF. The Latency observed via langfuse was that the single run took ~4.3 seconds end-to-end. The slowest hop is the **Extractor Agent (GPT-4o)** due to image processing and high-token-count inference.

- **Mitigation:** We implement asynchronous processing for ingestion. For CG operators, we surface "Processing" states in the UI so they can move to other tasks while the agent finishes its run.

Roadmap: 1 Week vs. 1 Day

If granted a full week for development, the following architectural upgrades would be prioritized to move from POC to Production-Ready:

- **Database Transition:** Migrate from SQLite to **Postgres** for multi-tenant isolation and reliability.
- **Dynamic Rule Retrieval:** Replace hardcoded rule dictionaries with a **RAG pipeline** to pull only relevant customer requirements into the Validator's context.
- **Cross-Document Validation:** Build a fourth agent to ensure consistency across the entire shipment packet (e.g., matching HS codes across BOL and Packing List).
- **Feedback Loops:** Implement a mechanism where CG overrides/edits flow back into the system to refine prompts and improve future extraction accuracy.